SEQUENCE 6

Functions: representations and reference functions

23 November–4 December 2026

6.1 Function, image and preimage

Key point · Definition. A real-valued function associates to each real number x of its domain D_f a unique real number $f(x)$, called the image of x. A preimage of y is a real number $x \in D_f$ such that $f(x) = y$. We write $f : x \mapsto f(x)$. A number may have no preimage, one, or several.

Example 1. Let $g(x) = x^2 - 4x + 32$ on $\mathbb{R}$. Calculate $g(-5)$. Let $h(x) = 3x + 5$; determine the preimage of 6 under h.

Example 2. For the square function, calculate the images of -3 and 3, then the preimages of 9, 0 and -1.

```
def g(x):
    return x*x - 4*x + 32
```

Example 3. Distinguish `g(-5)` from `x = -5`.

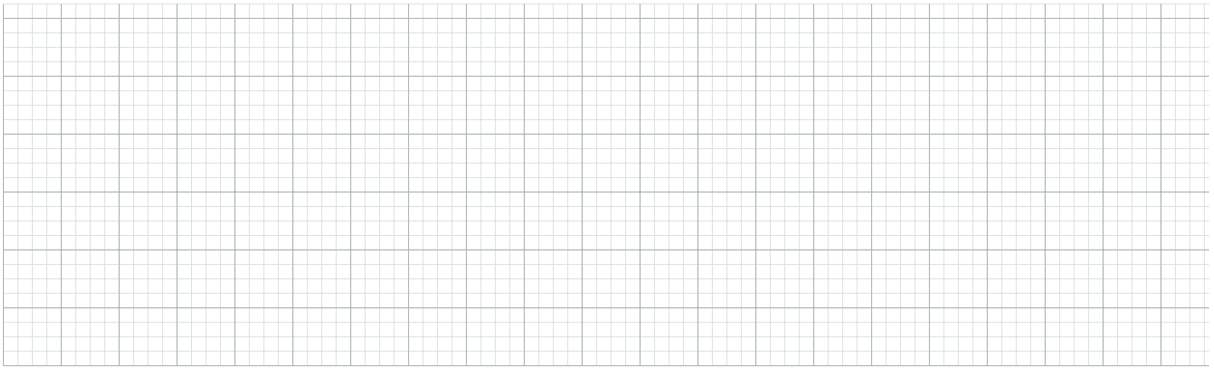
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6.2 Domain of definition

Key point · Conditions for existence. D_f contains the values for which the process defining f makes sense. A denominator must be nonzero; a number under a square root must be positive or zero. Also respect any domain imposed by the statement or the context.

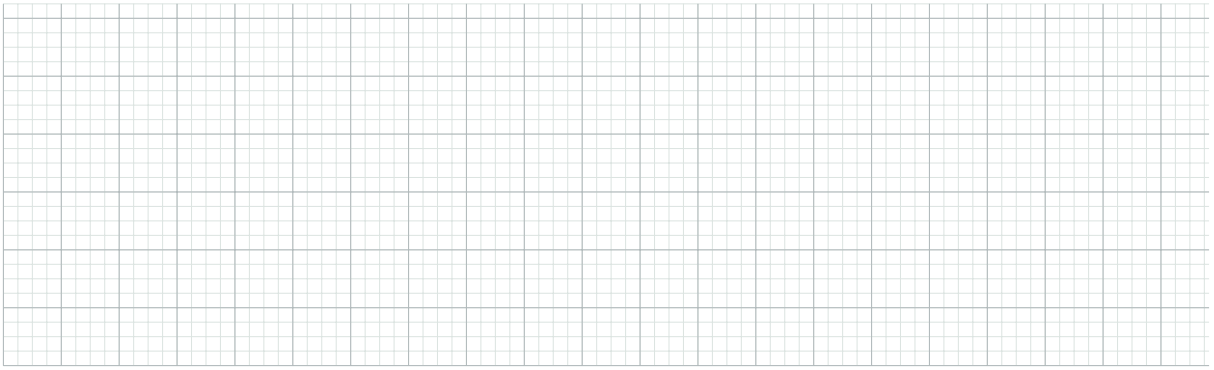
Example 4. Determine the domains of definition:

$$f(x) = 3x^2 - 5x + 2, \quad g(x) = \frac{2x+1}{x-4}, \quad h(x) = \frac{x}{x^2-9}, \quad k(x) = \sqrt{2x-6}.$$



Watch out. The square root of a negative number does not exist in $\mathbb{R}$; it is not "a negative square root" that is excluded.

Example 5 · Modelling. A rectangle of perimeter 20 m has width x m. Express its area and state the domain if both dimensions must be strictly positive.

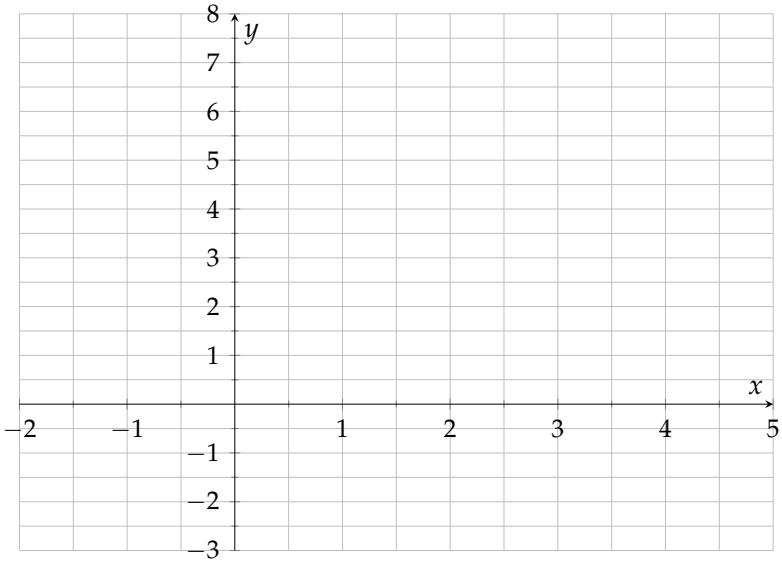


6.3 Table of values and representative curve

Key point · Definition. $\mathcal{C}_f$ is the set of points $(x, f(x))$, with $x \in D_f$. Thus $M(a,b) \in \mathcal{C}_f \iff a \in D_f$ and $b = f(a)$.

Example 6. On $[-1,4]$, $f(x) = x^2 - 2x - 1$. Complete the table, then draw the curve over the whole interval.

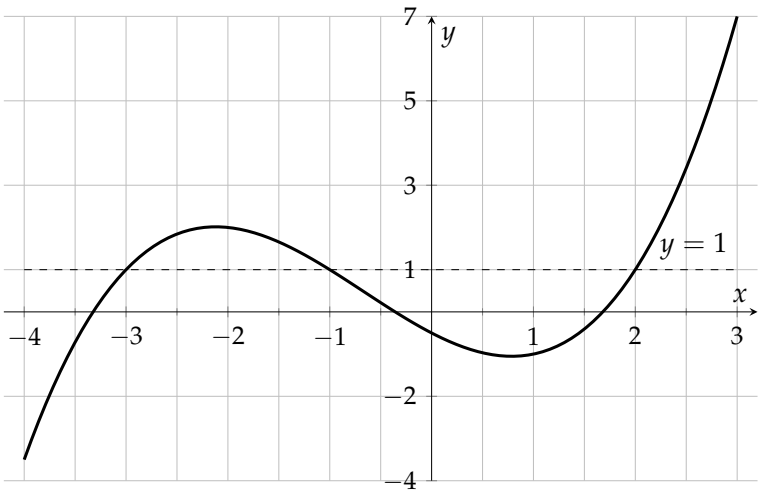
x	-1	-0.5	0	1	1.5	2	3	4
$f(x)$	.....	.....	-1	-2	.....	-1	.....	.....



6.4 Graphical solving of equations and inequalities

Key point · Method. The solutions of $f(x) = k$ are the abscissae of the intersections of $\mathcal{C}_f$ with $y = k$. For $f(x) \leq k$, read the abscissae where the curve is below this line, intersections included; with $<$, the intersections are excluded.

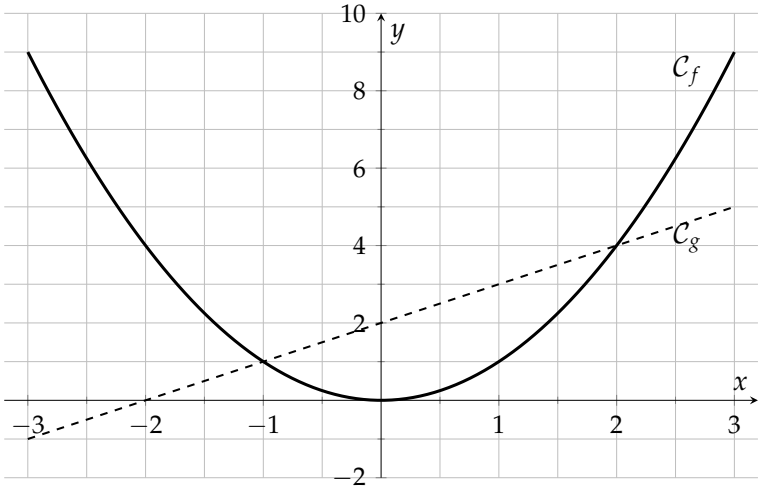
Example 7. The curve represents $f(x) = 1 + \frac{1}{4}(x+3)(x+1)(x-2)$ on $[-4, 3]$. Solve graphically $f(x) = 1$, $f(x) \leq 1$ and $f(x) > 1$.



6.5 Comparing two functions; reading a sign

Key point · Method. $f(x) = g(x)$ is read at the intersections of the curves. For $f(x) < g(x)$, $\mathcal{C}_f$ is strictly below $\mathcal{C}_g$ at the same abscissa. The sign of $f(x)$ is read relative to the x-axis.

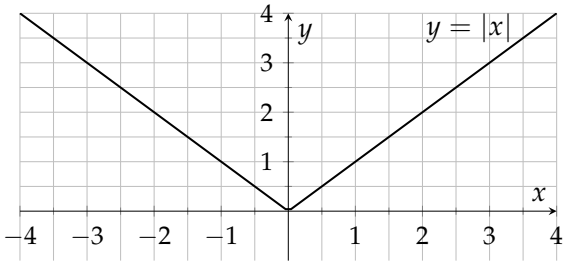
Example 8. On $[-3,3]$, $f(x) = x^2$ and $g(x) = x + 2$. Solve $f = g$, $f < g$, $f \geq g$ and determine the sign of g.



6.6 Affine functions and absolute value

Key point · Definitions. An affine function is $f(x) = mx + p$ on $\mathbb{R}$. It is linear if $p = 0$, constant if $m = 0$. For $m \neq 0$, it vanishes at $-p/m$; its sign is that of m to the right of this number. The function $x \mapsto |x|$ is defined on $\mathbb{R}$, positive or zero, and vanishes only at 0.

Example 9. Classify (several answers possible) as affine, linear, constant: $5x + 2$; $5x$; $7 + 2x - 7$; 6; $6(4x - 2)$; $6x + 5 - 6x$.



Key point · Equations and inequalities. If $k < 0$, $|x| = k$ and $|x| \leq k$ have no solution, and $|x| \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

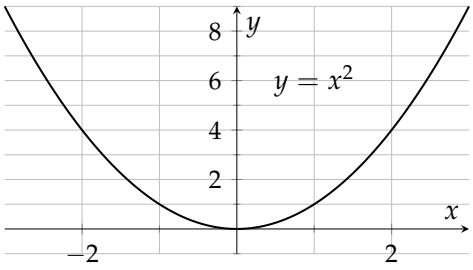
$$|x| = k \iff x \in \{-k, k\}, \quad |x| \leq k \iff x \in [-k, k],$$

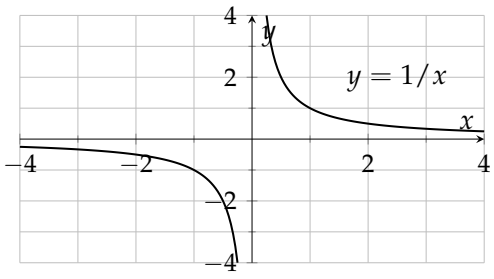
$$|x| \geq k \iff x \in (-\infty, -k] \cup [k, +\infty).$$

Example 10. Solve $|x| = 5$, $|x| = -3$, $|x| \leq 3$, $|x| \geq 4$.

6.7 Square and inverse functions

Key point · Definitions and sign. $x \mapsto x^2$ is defined on $\mathbb{R}$; its curve is a parabola. $x^2 \geq 0$, with equality only at 0. $x \mapsto 1/x$ is defined on $\mathbb{R}^*$; its curve is a hyperbola. $1/x$ has the sign of x and never vanishes.





Key point · Equations and inequalities with x^2. If $k < 0$, $x^2 = k$ and $x^2 \leq k$ have no solution; $x^2 \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

$$x^2 = k \iff x \in \{-\sqrt{k}, \sqrt{k}\}, \quad x^2 \leq k \iff x \in [-\sqrt{k}, \sqrt{k}],$$

$$x^2 \geq k \iff x \in (-\infty, -\sqrt{k}] \cup [\sqrt{k}, +\infty).$$

Example 11. Solve $x^2 = 4$, $x^2 = -1$, $x^2 \geq 4$, $x^2 \geq -1$, $1/x = 4$, $1/x = 0$.

6.8 Inequalities with the inverse function

Method. Respect $x \neq 0$ and read both branches of the hyperbola. Do not multiply an inequality by x without knowing its sign.

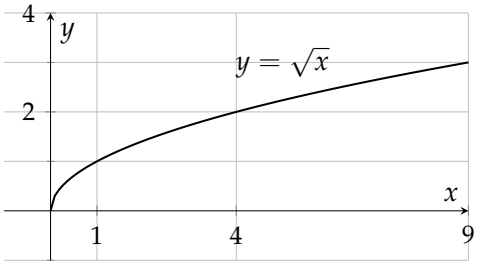
	$k < 0$	$k = 0$	$k > 0$
$1/x \leq k$	$[1/k, 0)$	$(-\infty, 0)$	$(-\infty, 0) \cup [1/k, +\infty)$
$1/x \geq k$	$(-\infty, 1/k] \cup (0, +\infty)$	$(0, +\infty)$	$(0, 1/k]$

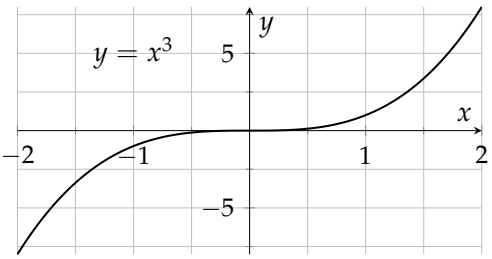
Example 12. Solve $1/x \geq 2$, $1/x \leq -1$, $1/x < 1/2$ and $1/x > -2$.

Example 13. A student writes: "$1/x \leq 2 \iff 1 \leq 2x$, so $x \geq 1/2$". What is missing?

6.9 Using square root and cube

Key point · Definitions. $x \mapsto \sqrt{x}$ is defined on $[0, +\infty)$ and takes values that are positive or zero. For $k \geq 0$, $\sqrt{x} = k \iff x = k^2$; for $k < 0$, no solution. $x \mapsto x^3$ is defined on $\mathbb{R}$. Every real number k has a unique preimage, written $\sqrt[3]{k}$.





Example 14. Solve $\sqrt{x} = 3$; $\sqrt{x} = -2$; $\sqrt{x} \leq 2$; $\sqrt{x} > 3$; $x^3 = -8$; $x^3 \leq 27$.

Example 15. Compare x^2 and x for $x = 1/2$, then $x = 2$. Is the square of a positive number always greater than the number itself?